Therefore

$$[\mathcal{L}_{H_1}, \mathcal{L}_{H_2}]f = -\{\{H_2, H_1\}, f\} = \{f, \{H_2, H_1\}\}.$$

This means that it is possible to relate the commutator of Lie derivatives of Hamiltonian vector fields to the Lie derivative of the Poisson bracket of the corresponding Hamiltonians, i.e.,

$$[\mathcal{L}_{H_1}, \mathcal{L}_{H_2}]f = \mathcal{L}_{\{H_1, H_2\}}f.$$

Because of this, we may interpret the first term in the expansion of the error as itself being derived from a Hamiltonian. This is a crucial observation which suggests an alternative perspective on the error expansion in splitting. The idea is to view the product of exponentials as an exponential:

$$e^{hA}e^{hB} = e^{h(A+B+hR)},$$

for some operator R. Expanding out the right hand side gives

$$\begin{aligned} e^{h(A+B+hR)} &= \mathrm{Id} + h(A + B + hR) \\ &\quad + \frac{h^2}{2}(A + B + hR)^2 + \dots \\ &= \mathrm{Id} + hA + hB + h^2R \\ &\quad + \frac{h^2}{2}(A^2 + AB + BA + B^2 \\ &\qquad + h^2R^2 + h(A + B)R + hR(A + B)) + \dots \end{aligned}$$

Comparing this to our expansion for $e^{hA}e^{hB}$, for agreement to $\mathcal{O}(h^2)$ we must have

$$\frac{1}{2}(2AB + A^2 + B^2) = R + \frac{1}{2}(A^2 + AB + BA + B^2) = \mathcal{O}(h),$$

or

$$R = AB - \frac{1}{2}(AB + BA) = \frac{1}{2}[A, B] + \mathcal{O}(h),$$

and from this it follows that

$$e^{h\mathcal{L}_{H_1}}e^{h\mathcal{L}_{H_2}} = e^{h\mathcal{L}_G},$$

where

$$G = H_1 + H_2 + \frac{h}{2}\{H_1, H_2\} + \mathcal{O}(h^2).$$